

Project Title: Synthesis and Evaluation of a $\text{CeO}_2\text{--MnO}_x$ Catalytic Ceramic Filter Membrane for Simultaneous Particulate Separation and Carbon Monoxide/VOC Oxidation

Focus Areas: Heterogeneous Catalytic Reaction Kinetics, Porous Media Transport, Flue-Gas Remediation

Target Application: Abatement of Brick Kiln Flue-Gas Emissions (CO , Soot, and Organic Pollutants)

1. Project Background and Research Objective

Traditional brick kilns produce high volumes of fly ash, soot, carbon monoxide (CO), and unburned volatile hydrocarbons. Conventional downstream cleanup methods decouple particulate filtration from gas scrubbing. This design causes rapid operational failure: standard packed adsorption beds foul within hours due to ash blinding, while fabric baghouses provide zero chemical destruction of toxic gaseous compounds.

The objective of this project is to develop and evaluate a bifunctional **catalytic filter membrane**. By anchoring a nanostructured transition metal oxide ($\text{CeO}_2\text{--MnO}_x$) directly onto heat-resistant woven fiberglass/ceramic substrate, the unit achieves:

1. **Physical surface-cake filtration** of solid particulates and ash at the outer face.
2. **Heterogeneous catalytic oxidation** of CO and gaseous organics as the flue gas permeates across the catalytic fiber matrix via the Mars–van Krevelen redox mechanism.

2. Chemical Mechanisms and Synthesis Route

2.1 Catalytic Mechanism (Mars–van Krevelen Cycle)

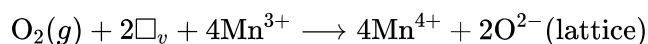
The oxidation of toxic CO and organics does not rely on permanent gas storage or adsorption capacity; it proceeds via continuous lattice oxygen donation and gas-phase replenishment:

1. **Lattice Oxygen Depletion (Surface Reduction):**



where $\square_v$ is an empty anionic oxygen vacancy in the catalyst lattice.

2. **Lattice Regeneration (Gas-Phase Reoxidation):**



Excess oxygen present in the kiln exhaust continuously reoxidizes reduced metal sites, preventing catalyst exhaustion.

3. **Catalytic Soot Burn-off:**

Surface-active oxygen species (O_2^- , O_2^{2-}) lower the carbon soot ignition temperature from over 600°C to the 300°C – 350°C range, preventing soot-tar accumulation on the outer face.

2.2 Material Synthesis Protocol

1. **Substrate Pre-treatment:** High-temperature woven E-glass or ceramic cloth coupons (25 mm or 47 mm diameter) are submerged in 10 wt% HNO_3 at 60°C for 45 minutes to chemically remove sizing agents, rinsed with deionized water, and dried at 110°C .
2. **Sol-Gel Precursor Synthesis:** Cerium(III) nitrate ($\text{Ce}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$) and manganese(II) nitrate ($\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$) are dissolved in an equimolar ratio in an ethanol-water solution. Citric acid monohydrate is added at a 1.2 : 1 molar ratio to total cations to prevent localized precipitation.
3. **Dip-Coating:** Pre-treated fabric coupons are dipped into the sol for 5 minutes, withdrawn at 1.5 cm/min, dried at 90°C for 1 hour, and repeated for 3 coating cycles to achieve an 8 – 12 wt% catalyst loading.
4. **Thermal Calcination:** The coated substrate is heated in air at a $2^\circ\text{C}/\text{min}$ ramp to 450°C and held for 3 hours to decompose the precursors into crystalline, high-surface-area $\text{CeO}_2\text{--MnO}_x$ nanoparticles.

3. Experimental Methodology

- **Reactor Configuration:** A tubular vertical quartz reactor clamped inside a programmable split-tube furnace (150°C – 450°C). The catalytic membrane disc is sealed across the gas path using graphite gaskets.
- **Gas Feed Simulation:** Calibrated mass flow controllers meter synthetic flue gas consisting of 1,000 ppm CO, 5% O₂, and balance N₂ through the membrane at controlled superficial face velocities (0.02 – 0.20 m/s).
- **Particulate Resistance Evaluation:** Pre-weighed carbon soot or fine fly ash is aerosolized and deposited on the upstream side to evaluate pressure drop changes across the membrane during operation.
- **Online Analytical Equipment:**
 - Differential pressure transmitter connected across upstream and downstream taps.
 - Non-dispersive infrared (NDIR) gas analyzer tracking effluent CO and CO₂ continuous concentrations.

4. End-of-Project Calculations and CRE Metrics

The following quantitative parameters will be calculated from the raw experimental data:

4.1 Fractional Conversion (X_{CO}) and Light-Off Profile

Fractional conversion is determined from inlet and outlet continuous concentrations:

$$X_{CO} = \frac{C_{CO, in} - C_{CO, out}}{C_{CO, in}}$$

- **Calculation:** Plot X_{CO} as a function of temperature (150°C – 450°C).
- **Deliverable:** Determine the characteristic light-off temperatures T_{50} (50% conversion) and T_{90} (90% conversion) to evaluate catalyst activity.

4.2 Apparent Reaction Rate ($-r'_{CO}$) and Kinetic Constants

At low conversion levels ($X_{CO} < 0.15$), the differential reactor model applies:

$$-r'_{CO} = \frac{F_{CO, in} \cdot X_{CO}}{W_{cat}}$$

where $F_{CO, in}$ is the molar inlet flow rate of CO (mol/s) and W_{cat} is the dry mass of catalyst deposited on the fabric (g).

- **Power-Law Form:** $-r'_{CO} = k_{app} C_{CO}^n$
- **Reaction Order (n):** Obtained by running isothermal experiments across multiple inlet concentrations (500 – 3,000 ppm) and calculating the slope of $\ln(-r'_{CO})$ versus $\ln(C_{CO})$.

4.3 Apparent Activation Energy (E_a)

The temperature dependence of the reaction rate constant is modeled via the Arrhenius equation:

$$k_{app} = k_0 \exp\left(-\frac{E_a}{RT}\right)$$

- **Calculation:** Plot $\ln(k_{app})$ versus $1/T$ (T in Kelvin).
- **Deliverable:** The apparent activation energy E_a (kJ/mol) is extracted directly from the slope:

$$\text{Slope} = -\frac{E_a}{R}$$

4.4 Mass Transfer and Internal Diffusion Verification

To confirm that recorded kinetic data represents true chemical reaction rather than intra-fiber diffusion limitations, evaluate the **Weisz-Prater Criterion** ($C_{W P}$):

$$C_{W P} = \frac{-r_{\text{CO, obs}} \cdot \rho_p \cdot R_p^2}{D_{\text{eff}} \cdot C_{\text{CO, surf}}}$$

where:

- $-r_{\text{CO, obs}}$ = observed reaction rate per unit mass of catalyst (mol/(kg · s))
- ρ_p = catalyst layer density (kg/m³)
- R_p = characteristic diffusion length / active layer thickness (m)
- D_{eff} = effective Knudsen/molecular diffusion coefficient (m²/s)
- $C_{\text{CO, surf}}$ = concentration of CO at the external fiber surface (mol/m³)
- **Criterion:** If $C_{W P} \ll 1$, internal pore diffusion resistance is negligible, verifying intrinsic chemical kinetic control.

4.5 Membrane Permeability and Cake Resistance

Pressure drop across the clean and soot-laden fabric is evaluated using the combined Darcy-Forchheimer and filtration resistance equations:

$$\Delta P = \Delta P_{\text{membrane}} + \Delta P_{\text{cake}} = v_0 \left(\frac{\mu \cdot L_m}{\kappa_m} + \mu \cdot r_c \cdot w_c \right)$$

where:

- v_0 = superficial gas velocity (m/s)
- μ = flue-gas dynamic viscosity (Pa · s)
- L_m = membrane thickness (m)
- w_c = soot/ash mass deposited per unit area (kg/m²)
- **Deliverables to calculate:**
 1. **Clean Permeability** (κ_m , m²): Calculated from the linear slope of ΔP versus v_0 prior to ash introduction.
 2. **Specific Cake Resistance** (r_c , m/kg): Calculated by plotting $(\Delta P/v_0)$ versus accumulated ash mass w_c .

5. Summary Table of Deliverables and Success Metrics

Measured / Calculated Metric	Mathematical Formula / Method	Acceptance / Benchmark Criterion
Light-Off Temperature (T_{50})	T at which $X_{\text{CO}} = 0.50$	$T_{50} \leq 250^{\circ}\text{C}$
Apparent Activation Energy (E_a)	Arrhenius slope: $-E_a/R$	Significant reduction vs. uncatalyzed gas oxidation ($> 150 \text{ kJ/mol} \rightarrow < 60 \text{ kJ/mol}$)
Internal Diffusion Limitation ($C_{W P}$)	Weisz-Prater parameter	$C_{W P} < 0.3$ (Intrinsic kinetic regime validated)

Measured / Calculated Metric	Mathematical Formula / Method	Acceptance / Benchmark Criterion
Clean Filter Permeability (κ_m)	Darcy slope: $\kappa_m = \mu L_m / (\Delta P / v_0)$	$\Delta P < 150$ Pa at typical baghouse velocity (0.05 m/s)
Catalyst Adhesion Durability	Mass loss after 15 min sonication	Catalyst washcoat loss < 3 wt%